

Effect of Different Types of Occlusion on the Supporting Structures of Mandibular Overdenture Retained by Two Splinted Mini-Implants

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Abstract

Objective: To evaluate and compare the effect of lingualized occlusion and monoplane occlusion on the peri-implant bone level of mandibular overdenture retained by two splinted mini-implants using Cone Beam Computed Tomography(CBCT).

Methodology: Fourteen completely edentulous patients were divided randomly into two equal groups: Group I received splinted Mini-implant retaining mandibular overdentures with their occlusal scheme set according to the lingualized concept. Group II received splinted Mini-implant retaining mandibular overdentures with their occlusal scheme set according to the monoplane concept. The peri-implant marginal bone loss was evaluated at zero, six and twelve months from the time of loading using CBCT.

Findings: The crestal bone loss around mini-implants in Monoplane group is higher than Lingualized group with no significant difference.

Conclusion: In case of splinted mini-implants retaining mandibular overdenture, type of occlusion has no effect on the supporting structures.

Keywords: Mandibular Overdenture - Mini-Implants-Occlusion.

Introduction

Implant-retained overdentures are recommended for completely edentulous patients to create denture stability and to avoid problems with denture wearing as retention and satisfaction.⁽¹⁾

Splinted mini-implants supporting a mandibular overdenture showed less marginal bone loss compared with nonsplinted mini-implants.⁽²⁾

Occlusion is one of the critical factors for implant longevity. ⁽³⁾Two occlusal schemes were selected, the lingualized and the monoplane occlusion. These schemes decrease occlusal forces falling on the supporting structures, lingualized occlusion was considered as the choice for complete dentures as well as tooth supported overdentures. It provides mechanical freedom in centric which decrease horizontal (destructive) stresses and controls leverage induced by eccentric tooth contact.^(4, 5)

Monoplane occlusion can adapt to the slight discrepancies between centric relation and centric occlusion that usually occur due to processing errors of the acrylic resin denture base and settling of the denture after wear. Elimination of cusps of posterior teeth decreases the horizontal forces transferred to the supporting structures. ^(4, 5)

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Thus the objective was evaluation of peri-implant marginal bone loss of lingualized occlusion and

monoplane occlusion in two splinted mini-implants retaining mandibular overdenture using CBCT.

Materials and Method

I-Patient Selection: Fourteen completely edentulous patients with age ranging between 50-65 years were selected randomly from the Outpatient Clinic, Faculty of Dentistry, Ain Shams University. Patients had residual bone and mucosa free from any pathological signs or bony undercuts with adequate interarch space and normal ridge relationship. The radiographic examination revealed adequate bone width and height in the interforaminal region to accept mini implants of 2.5mm diameters and 10mm lengths.

Patients were free from any systemic diseases that might affect healing, complicate surgical procedures or contribute to bone resorption. Patients with inadequate oral hygiene or with history of head and neck radiotherapy or chemotherapy were excluded.

All patients agreed to be a part of this study after being informed about the procedure and agreed to undergo a follow-up period for one year.

Grouping of the patients:

Patients were divided randomly into two equal groups (seven patients each): Both were rehabilitated by a mucosa supported maxillary complete denture and mandibular overdenture retained by two splinted mini-implants with different occlusal schemes:

Group (I): Rehabilitated with lingualized occlusion.

Group (II): Rehabilitated with monoplane occlusion.

II-Complete Denture Construction: Primary alginate impressions were made then secondary impressions using zinc oxide-eugenol impression material to produce occlusion blocks.

For Group I Patients: A face bow record was made to mount the maxillary cast on a semi adjustable articulator. Centric occluding relation record was made to mount the mandibular cast in the correct vertical dimension following the check bite technique. Protrusive jaw relation record was used to adjust the horizontal condylar guidance of the articulator. The lateral condylar guidance was adjusted according to Hanau's formula ($L = H/8 + 12$).

Modified cross-linked acrylic teeth were arranged

following the lingualized occlusion. Anterior teeth were arranged according to aesthetics, phonetics and lip support. Lingual cusps of mandibular posterior teeth were placed medial to the lines drawn from the tip of the canine to both sides of the retromolar pad. Mandibular posterior teeth were positioned with no transverse inclination. The interlocking transverse ridges of lower posterior teeth were removed. Compensating curve must be created in harmony with the incisal guidance anteriorly and the condylar guidance posteriorly. Maxillary posterior teeth were arranged with their lingual cusps occluding in the modified central fossae of the mandibular posterior teeth and the buccal cusps of maxillary posterior teeth were reduced about 1 mm to eliminate buccal cusp contact both in centric and working jaw positions.

For Group II Patients: A face bow record was made to mount the maxillary cast on a mean value articulator. Centric occluding relation was recorded at the predetermined vertical dimension of occlusion following the check bite technique. Upper and lower casts were mounted on mean value articulator.

Modified cross-linked acrylic teeth were used after flattening the cusps and arranged following the monoplane occlusion. Anterior teeth were arranged without any overlap. The teeth were flattened and arranged anteroposteriorly so as to be parallel to the plane of the denture foundation. Mediolaterally, the teeth were positioned flat with no medial or lateral inclination. The lower second molar was kept out of occlusion to direct the forces to the bicuspid molar region.

For both groups, the waxed up dentures were then tried in the patient's mouth and then processed following the long curing cycle.

Laboratory remounting was carried out before finishing and polishing of the denture. Clinical remounting was done for final occlusal adjustments.

At the delivery appointment, final occlusal adjustment were done and dentures were delivered to the patients to get used to it and post insertion instructions were given to the patients.

Construction of surgical stent: Patient's lower denture was duplicated to produce radiographic stent. Estimation of bone width and height to reveal about 4mm width and 12mm height then the radiographic stent was converted to surgical stent by drilling channels in the planned mini-implant positions.

Surgical Procedures: Bilateral mental nerve block and Field block anesthesia were given. Crestal incision was done at the canine-premolar region and reflected. The surgical stent was seated in the patient's mouth, Initial penetration through the cortex of the bone was made by a Point Drill through the hole of the Surgical stent which represented the planned position of the implant. The pilot hole was widened with diameter 1.3 drill then with diameter 1.8 drill until a depth of 10mm was reached.

The screw- type square head (cement) mini implant was inserted to its full length with all the threads were placed into the bone using ratchet driver and torque ratchet wrench. The second mini-implant on the opposite side was placed with the same steps then the flaps were sutured.

The abutments of the square head mini-implants are marked at two different heights, the abutments of all patients were trimmed to the lowest mark as it allowed fixed abutment heights.

Steps of impression making for bar construction: Patients were recalled one week later. The plastic Transfer copings were placed over the abutments of the mini- implants in the patient mouth then rubber base impression was taken. Implant analogues were connected to the transfer copings in the impression material.

Steps of the Laboratory work for the construction of the bar: Stone cast was poured with implant analogue in its place representing the position of the mini-implants in patient's mouth.

Wax pattern copings were fabricated over the abutments of the implant analogue.

Prefabricated plastic pattern of the bar was trimmed to a suitable length to connect the copings.

The bar-coping pattern was removed from the abutments of the implant analogues and casted.

Steps of loading of the mini-implants: One week later, the bar was tried in the patient's mouth, any adjustments were done then it was cemented using Self –adhesive resin cement. The position of the bar was detected in the fitting surface of the lower denture and area marked by the bar removed using large round bur. The denture was inserted in the patient's mouth to

ensure that it was fully seated. Two Plastic bar clips were attached to the bar. The undercuts were blocked out using a putty rubber base impression material.



Fig. (1): The clips were attached to the cemented bar.

The relieved denture must be tried again to ensure enough space for bar with the connected clips and the rubber base impression material which blocked out the undercuts. Self-cured acrylic resin (Dough stage) placed in the relieved area in the fitting surface of the lower denture. The denture was inserted in the patient mouth. The patient was asked to bite with the maximum intercuspation then the denture was removed with the clips picked-up in the fitting surface of the lower denture.

Final occlusal adjustments were done using T-Scan device to ensure accurate distribution of the forces on the teeth for each group and dentures were delivered to the patients with the clips picked-up in the fitting surface of the lower denture and post insertion instructions were given to the patients.

Follow up visits were scheduled six, twelve months after performing the needed clinical and radiographic evaluation of the mini-implants.

Radiographic Evaluation: CBCT was done at the time of the mini-implant loading as base line image (zero), six months, and twelve months after mini-implants loading.

The mini-implants were evaluated using the linear measurement system supplied by the cone beam computed tomography I CAT Vision



Fig. (2): The MPR screen showing the mini-implant appearing clearly in both coronal and sagittal planes.

All the linear measurements were measured at the MPR screen (Multi-Planar reconstruction).

The mean values of all aspects (buccal, lingual, mesial and distal) were calculated.

Results were collected, tabulated, and statistically analyzed using one-way ANOVA of repeated measures for intragroup comparisons followed by bonferroni

post hoc test and independent t-test for Intergroup comparisons to evaluate effect of different types of occlusion on the marginal bone loss around two splinted mini-implants retaining mandibular overdenture.

The significance level was set at $P \leq 0.05$ within all tests. Statistical analysis was performed with IBM SPSS Statistics Version 25 for Windows.

Findings:

Crestal Bone Loss:

A-Intragroup Comparison:

Table (1): Mean and Standard deviation (SD) and p- values for differences in crestal bone loss (mm) values at different time intervals in each groups

Intervals	Crestal bone loss	
	Lingualized occlusion	Monoplane occlusion
Baseline-6 months	0.64±0.13 ^B	0.72±0.09 ^B
6 months-12 months	0.20±0.03 ^C	0.22±0.05 ^C
Baseline-12 months	0.84±0.14 ^A	0.94±0.12 ^A
p-value	<0.001*	<0.001*

Different superscript letters indicate a statistically significant difference within the same vertical column *; significant ($p \leq 0.05$) ns; non-significant ($p > 0.05$).

Table 1 showed that For both Lingualized and Monoplane occlusion groups, the results show bone loss for each group. This decrease was statistically significant

throughout the study period where the greatest decrease was found in baseline-12 months and the least was found in 6-12 months. However by using bonferroni post hoc test, there was a statistically significant difference between baseline-6 months and 6-12 months, between baseline-6 months and baseline-12 months, between 6-12 months and baseline-12 months.

B-Intergroup Comparison:**Table (2): Mean and Standard deviation (SD) and p- values for differences in crestal bone loss (mm) values at different time intervals between groups**

Intervals	Crestal bone loss		p-value
	Lingualized occlusion	Monoplane occlusion	
Baseline-6 months	0.64±0.13	0.72±0.09	0.216ns
6 months-12 months	0.20±0.03	0.22±0.05	0.347ns
Baseline-12 months	0.84±0.14	0.94±0.12	0.182ns

*Significant ($p \leq 0.05$) ns; non-significant ($p > 0.05$)

Table 2 showed higher bone loss in group II (Monoplane occlusion groups) than group I (Lingualized occlusion group). However, this difference was statistically insignificant throughout the study period.

Discussion

The patients were selected and examined to eliminate any factor or habit that may adversely affect the results. This was done using comprehensive medical history, clinical examination followed by the laboratory investigation.^(6,7)

Mini-implant retained mandibular over-dentures present a simple and reliable solution for denture stability and retention problems.⁽⁸⁾

Two protective occlusal schemes were selected, the lingualized and the monoplane occlusion. The lingualized occlusion provides mechanical freedom in centric which decrease horizontal (destructive) stresses and controls leverage induced by eccentric tooth contact. The second occlusal scheme was the monoplane occlusion, it can adapt to the slight discrepancies between centric relation and centric occlusion that usually occur due to processing errors of the acrylic resin denture base and settling of the denture after wear.⁽⁹⁾

Remounting techniques were carried out to assess and correct any occlusal discrepancies, which might lead to lateral forces on the implants.⁽¹⁰⁾

The selected implant was 2.5 mm in diameter, the width of the ridge in the proposed implant site was at least 4 mm measured by CBCT. From a biomechanical point of view, it is well documented that a residual plate that is less than 2 mm in thickness could have an adverse affect on bone stress levels and crestal bone maintenance.⁽¹¹⁾

Bar attachment has been used to splint the two mini-implant in this study. Bar/Clip attachment helps with abutments splinting, offers high retention capacity and minimizes prosthesis movement during function.⁽¹²⁾

The linear measurements were done using a regular CBCT radiographic examination to evaluate the bone loss, the period of 6 months could be sufficient for a measurable bone loss around the mini-implants also, a whole period of 12 months could be used to evaluate the acceptable bone loss around the mini-implants.⁽¹³⁾

A significant decrease in the crestal bone height was observed during the follow-up periods around the mini-implants in both groups due to immediate bone reaction after insertion of the prosthesis beside healing process after the implant placement, progressive reduction of bone height till the end of the study period might be due to the cumulative effect of mechanical factors acting on the implants as loading and forces of mastication.⁽¹⁵⁾

The mean values of crestal bone loss in this study within the 12 months follow up period were in agreement with the success criteria within the accepted range to happen within the first year of implant placement.

The minimal bone loss values recorded in this study may be attributed to the type of occlusal schemes used (monoplane and lingualized) because they are both considered “non-traumatic protective” occlusal schemes.

The difference between the two groups was statistically insignificant due to the favorable force transmission offered by both occlusal schemes.

Both were considered successful in preserving the crestal bone around the mini-implants meanwhile at the end of the study period, The results show that the total bone loss was slightly higher in Monoplane group than in lingualized group.

This may be attributed to increased cutting efficiency and less number of strokes to smash food which decrease load on the underlying supporting structures in dentures of lingualized group.⁽¹⁶⁾

Conclusion

In case of splinted mini-implants retaining mandibular overdenture, type of occlusion has no effect on the supporting structures.

Conflict of Interest: Nil.

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Ethical Clearance: Taken from Faculty of Dentistry Ain Shams University, Research Ethics Committee (FDASU-REC)

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